B. Maternal and fetal assessment

B.1: Maternal assessment

Background

Hypertensive disorders of pregnancy are important causes of maternal and perinatal morbidity and mortality, with approximately a quarter of maternal deaths and near misses estimated to be due to pre-eclampsia and eclampsia *(9)*. Antenatal screening for pre-eclampsia is an essential part of good ANC. It is routinely performed by measuring maternal blood pressure and checking for proteinuria at each ANC contact and, upon detection of pre-eclampsia, specific management is required to prevent eclampsia and other poor maternal and perinatal outcomes *(57)*. The GDG did not evaluate evidence or make a recommendation on this procedure, therefore, which it considers to be an essential component of Good Clinical Practice in ANC.

As part of the ANC guideline development, specifically in relation to maternal assessment, the GDG considered evidence and other relevant information on interventions to detect the following conditions in pregnancy:

- **Anaemia:** Defined as a blood haemoglobin (Hb) concentration below 110 g/L, anaemia is the world's second leading cause of disability, and one of the most serious global public health problems, with the global prevalence of anaemia among pregnant women at about 38% *(33)*. Clinical assessment (inspection of the conjunctiva for pallor) is a common method of detecting anaemia but has been shown to be quite inaccurate. In HICs, performing a full blood count, which quantifies the blood Hb level, is part of routine ANC *(81)*. However, this and other available tests may be expensive, complex or impractical for use in rural or LMIC settings. A low-cost and reliable method of detecting anaemia is therefore needed for places with no or limited access to laboratory facilities. WHO developed the haemoglobin colour scale, a low-cost method that is performed by placing a drop of undiluted blood on specially made chromatography paper and matching it against a range of colours representing different Hb values in 20 g/L increments *(82)*. With haemoglobinometer tests, undiluted blood is placed directly into a microcuvette, which is inserted into the haemoglobinometer (or photometer) to produce a reading *(82)*.

- **Asymptomatic bacteriuria (ASB):** ASB is a common urinary tract condition that is associated with an increased risk of urinary tract infections (cystitis and pyelonephritis) in pregnant women. Escherichia coli is associated with up to 80% of isolates; other pathogens include *Klebsiella* species, *Proteus mirabilis* and group B streptococcus (GBS) *(83)*. Methods for diagnosing ASB include midstream urine culture (the gold standard), Gram stain and urine dipstick tests. A urine culture can take up to seven days to get a result, with the threshold for diagnosis usually defined as the presence of 10^5 colony-forming units (cfu)/mL of a single organism *(84)*. The Gram stain test uses colour stains (crystal violet and safrinin O) to exaggerate and distinguish between Gram-positive (purple) and Gram-negative (red) organisms on a prepared glass slide. Urine dipsticks test for nitrites, which are not found in normal urine, and leucocytes, which are identified by a reaction with leucocyte esterase, to identify the presence of bacteria and pus in the urine, respectively. ASB is associated with an increased risk of preterm birth; once detected it is, therefore, usually actively managed with antibiotics (see also Recommendation C.1, in section C: Preventive measures).

- **Intimate partner violence (IPV):** IPV, defined as any behaviour within an intimate relationship that causes physical, psychological or sexual harm to those in the relationship, is now recognized as a global public health issue. Worldwide, almost one third of all women who have been in a relationship have experienced physical and/or sexual violence by their intimate partner *(85)*. Emotional abuse (being humiliated, insulted, intimidated and subjected to controlling behaviours such as not being permitted to see friends or family) also adversely impacts the health of individuals *(85)*.

IPV is associated with chronic problems in women, including poor reproductive health (e.g. a history of STIs including HIV, unintended pregnancy, abortion and/or miscarriage), depression, substance use and other mental health problems *(85)*. During pregnancy, IPV is a potentially preventable risk factor for various adverse outcomes, including maternal and fetal death. Clinical enquiry about IPV aims to identify women who have experienced or are experiencing IPV, in order to offer interventions leading to improved outcomes. Some governments and professional organizations recommend screening all women for IPV rather than asking only women with symptoms *(86)*.

Women's values

A scoping review of what women want from ANC and what outcomes they value informed the ANC guideline *(13)*. Evidence showed that women from high-, medium- and low-resource settings valued having a positive pregnancy experience. Within the context of maternal and fetal assessment, women valued the opportunity to receive screening and tests to optimize their health and that of their baby as long as individual procedures were explained to them clearly and administered by knowledgeable, supportive and respectful health-care practitioners (high confidence in the evidence).

In addition to GDG recommendations on the above, recommendations on diagnosing gestational diabetes mellitus (GDM) and screening for tobacco smoking, alcohol and substance abuse, TB and HIV infection have been integrated into this chapter from the respective existing WHO guidance on these conditions.

B.1.1: Anaemia

RECOMMENDATION B.1.1: Full blood count testing is the recommended method for diagnosing anaemia during pregnancy. In settings where full blood count testing is not available, on-site haemoglobin testing with a haemoglobinometer is recommended over the use of the haemoglobin colour scale as the method for diagnosing anaemia in pregnancy.
(Context-specific recommendation)

Remarks

- The GDG agreed that the high recurrent costs of Hb testing with haemoglobinometers might reduce the feasibility of this method in some low-resource settings, in which case the WHO haemoglobin colour scale method may be used.
- Other low-technology on-site methods for detecting anaemia need development and/or investigation.

Summary of evidence and considerations

Test accuracy of on-site Hb testing with haemoglobinometer and haemoglobin colour scale (HCS) methods to detect anaemia (EB Table B.1.1)

The evidence was derived from a test accuracy review conducted to support the ANC guideline *(81)*. Only one study (671 women) contributed data *(87)*. The study, conducted in Malawi, assessed the test accuracy of on-site Hb testing with a haemoglobinometer (HemoCue®) and the HCS method in comparison to a full blood count test performed by an electronic counter (Coulter counter), the reference standard.

Moderate-certainty evidence shows that the sensitivity and specificity of the haemoglobinometer test in detecting anaemia (Hb < 110 g/L) are approximately 0.85 (95% CI: 0.79–0.90) and 0.80 (95% CI: 0.76–0.83), respectively, while the sensitivity and specificity of the HCS method are lower at approximately 0.75 (95% CI: 0.71–0.80) and 0.47 (95% CI: 0.41–0.53), respectively.

For severe anaemia (defined in the study as Hb < 60 g/L), moderate-certainty evidence shows that the sensitivity and specificity of the haemoglobinometer test are approximately 0.83 (95% CI: 0.44–0.97) and 0.99 (95% CI: 0.98–1.00), respectively, while for the HCS method they are approximately 0.50 (95% CI: 0.15–0.85) and 0.98 (95% CI: 0.97–0.99), respectively.

Additional considerations

- In absolute numbers, the data mean that in settings with an anaemia prevalence of 38%, the haemoglobinometer test will probably miss about 57 anaemic women (95% CI: 38–80) out of every 1000 women tested, whereas the HCS method will probably miss about 95 anaemic women (95% CI: 76–110) out of every 1000 women tested. For populations with a severe anaemia prevalence of 5%, the haemoglobinometer test will probably miss about nine women with severe anaemia (95% CI: 2–27) out of every 1000 women tested, whereas the HCS method will probably miss about 25 women with severe anaemia (95% CI: 3–43) out of every 1000 women tested.
- The main limitation of the evidence is the low number of women identified with severe anaemia, which affects the precision of the estimates. However, the evidence suggests that the haemoglobinometer test is probably more accurate than the HCS method. As there are no direct comparisons in test accuracy studies and, as confidence intervals for sensitivity and specificity of the two methods overlap, there is some uncertainty about the relative accuracy of these tests.
- The review also evaluated the test accuracy of clinical assessment (4 studies, 1853 women), giving a sensitivity for clinical assessment of 0.64 (95% CI: 22–94) and a specificity of 0.63 (95% CI: 23–91) for detecting anaemia (Hb < 110 g/L). Thus, the HCS method might be more sensitive but less specific than clinical assessment.
- In settings where iron supplementation is routinely used by pregnant women, the consequence of missing women with severe anaemia is more serious than that of missing women with mild or moderate anaemia, as women with severe anaemia usually require additional treatment. Therefore, the accuracy of on-site Hb tests to detect severe anaemia in pregnancy is probably more important than the ability to detect Hb below 110 g/L.
- A study of various Hb testing methods in Malawi found the haemoglobinometer method to be the most user-friendly method *(82)*.

Values

Please see "Women's values" in section 3.B.1: Maternal assessment: Background (p. 41).

Resources

Any health-care provider can perform both the haemoglobinometer and HCS methods after minimal training. The haemoglobinometer and HCS methods have been estimated to cost approximately US$ 0.75 and US$ 0.12 per test, respectively *(82)*. Both methods require needles for finger pricks, cotton balls, gloves and Sterets® skin cleansing swabs; however, the higher costs associated with haemoglobinometer tests are mainly due to supplies (cuvettes and controls), equipment costs and maintenance.

Equity

The highest prevalence of maternal anaemia occurs in Africa and South-East Asia, where parasitic infections are major contributory factors *(33)*. Anaemia increases perinatal risks for mothers and newborns and contributes to preventable mortality. Accurate, low-cost, simple-to-use tests to detect anaemia might help to address health inequalities by improving the detection and subsequent management of women with anaemia, particularly severe anaemia, in low-resource settings.

Acceptability

Qualitative evidence from a variety of settings indicates that women generally appreciate clinical tests that support their well-being during pregnancy (moderate confidence in the evidence) *(22)*. However, evidence from LMICs indicates that where there are likely to be additional costs associated with tests, or where the recommended interventions are unavailable because of resource constraints, women may be less likely to engage with ANC services (high confidence in the evidence).

Feasibility

Qualitative evidence from providers in various LMICs indicates that a lack of resources, both in terms of the availability of the diagnostic equipment and potential treatments, as well as the lack of suitably trained staff to deliver the service, may limit implementation of recommended interventions (high confidence in the evidence) *(45)*.

B.1.2: Asymptomatic bacteriuria (ASB)

RECOMMENDATION B.1.2: Midstream urine culture is the recommended method for diagnosing asymptomatic bacteriuria (ASB) in pregnancy. In settings where urine culture is not available, on-site midstream urine Gram-staining is recommended over the use of dipstick tests as the method for diagnosing ASB in pregnancy. *(Context-specific recommendation)*

Remarks

- This recommendation should be considered alongside Recommendation C.1 on ASB treatment (see section C: Preventive measures).
- The GDG agreed that the higher resource costs associated with Gram stain testing might reduce the feasibility of this method in low-resource settings, in which case, dipstick tests may be used.
- The GDG agreed that ASB is a priority research topic, given its association with preterm birth and the uncertainty around urine testing and treatment in settings with different levels of ASB prevalence. Specifically, studies are needed that compare on-site testing and treatment versus testing plus confirmation of test with treatment on confirmatory culture, to explore health and other relevant outcomes, including acceptability, feasibility and antimicrobial resistance. In addition, better on-site tests need to be developed to improve accuracy and feasibility of testing and to reduce overtreatment of ASB. Research is also needed to determine the prevalence of ASB at which targeted testing and treatment rather than universal testing and treatment might be effective.

Summary of evidence and considerations

Test accuracy of on-site urine Gram staining and dipsticks to detect ASB (EB Table B.1.2)

The evidence was derived from a test accuracy review of on-site urine tests conducted to support the ANC guideline *(88)*. Four studies (1904 pregnant women) contributed data on urine Gram staining and eight studies (5690 pregnant women) contributed data on urine dipsticks. Most of the studies were conducted in LMICs. The average prevalence of ASB in the studies was 8%. A Gram stain was positive if one or more bacteria were detected per oil-immersed field, and a dipstick test was positive if it detected either nitrites or leucocytes. The reference standard used was urine culture with a threshold of 10^5 cfu/mL.

However, the certainty of the evidence on the accuracy of both Gram stain tests and dipstick tests is very low, with pooled sensitivity and specificity of the Gram stain test estimated at 0.86 (95% CI: 0.80–0.91) and 0.97 (95% CI: 0.93–0.99), respectively, and pooled sensitivity and specificity for urine dipsticks estimated at 0.73 (95% CI: 0.59–0.83) and 0.89 (95% CI: 0.79–0.94), respectively. A positive nitrite test alone on dipsticks was found to be less sensitive but more specific than when urine leucocytes were also considered.

Additional considerations

- A high level of accuracy in detecting ASB is important to avoid treating women unnecessarily, particularly in view of increasing antimicrobial resistance. Based on the uncertain evidence above, and assuming a prevalence of ASB of 9%, there would be 18 and 118 false-positive tests per 1000 women tested with Gram stain and dipstick tests, respectively. This suggests that, in settings where pregnant women are treated for ASB, dipstick diagnosis of ASB might lead to many women receiving unnecessary treatment.
- Dipstick tests are multi-test strips that, in addition to testing for nitrites and leucocytes, may also include detection of urine protein and glucose. However, the accuracy of dipsticks to detect conditions associated with proteinuria (pre-eclampsia) and glycosuria (diabetes mellitus) is considered to be low.

Values

Please see “Women’s values” in section 3.B.1: Maternal assessment: Background (p. 41).

Resources

Dipsticks are relatively low cost compared with the Gram stain test, as the latter requires trained staff and laboratory equipment and supplies (microscope, glass slides, reagents, Bunsen burner or slide warmer). Gram stain tests take longer to perform and to produce results than urine dipstick tests (10–30 minutes vs 60 seconds).

Equity

Preterm birth is the leading cause of neonatal death worldwide, with most deaths occurring in LMICs. Timely diagnosis and treatment of risk factors associated with preterm birth might therefore help to address health inequalities.

Acceptability

Qualitative evidence from a range of settings suggests that women view ANC as a source of knowledge, information and clinical expertise and that they generally appreciate the tests and advice they are offered (high confidence in the evidence) *(22)*. However, engagement with ANC services may be limited if tests and procedures are not explained properly or when women feel their beliefs and traditions are being overlooked or ignored by health-care professionals. In addition, if the Gram stain test is associated with long waiting times at ANC or having to return for test results, this may be less acceptable to women, as it might have additional cost and convenience implications for them (high confidence in the evidence). Health professionals are likely to prefer the dipstick test as it is associated with less effort (no need to label samples for laboratory assessment, perform tests or schedule follow-up visits to provide the results) and might provide additional information pertaining to other conditions (pre-eclampsia and diabetes mellitus) (high confidence in the evidence).

Feasibility

Qualitative evidence indicates that, in some LMIC settings, the lack of diagnostic equipment at ANC facilities discourages women from attending, and that providers often do not have the diagnostic equipment, supplies or skills to perform tests (high confidence in the evidence) *(45)*. Therefore, urine dipstick tests, which are cheaper and easy to perform, might be more feasible in low-resource settings.

B.1.3: Intimate partner violence (IPV)

RECOMMENDATION B.1.3: Clinical enquiry about the possibility of intimate partner violence (IPV) should be strongly considered at antenatal care visits when assessing conditions that may be caused or complicated by IPV in order to improve clinical diagnosis and subsequent care, where there is the capacity to provide a supportive response (including referral where appropriate) and where the WHO minimum requirements are met.[a] *(Context-specific recommendation)*

Remarks

- This recommendation is consistent with the 2013 publication *Responding to intimate partner violence and sexual violence against women: WHO clinical and policy guidelines (86)*. The evidence on clinical enquiry was indirect (strong recommendation) and the evidence on universal screening was judged as being of low to moderate quality (conditional recommendation).
- "Universal screening" or "routine enquiry" (i.e. asking all women at all health-care encounters) about IPV is not recommended. However, the WHO guidelines identify ANC as a setting where routine enquiry could be implemented if providers are well trained on a first-line response and minimum requirements are met *(86)*.
- Examples of conditions during pregnancy that may be caused or complicated by IPV include *(86)*:
  - traumatic injury, particularly if repeated and with vague or implausible explanations;
  - intrusive partner or husband present at consultations;
  - adverse reproductive outcomes, including multiple unintended pregnancies and/or terminations, delay in seeking ANC, adverse birth outcomes, repeated STIs;
  - unexplained or repeated genitourinary symptoms;
  - symptoms of depression and anxiety;
  - alcohol and other substance use;
  - self-harm, suicidality, symptoms of depression and anxiety.
- The GDG agreed that, despite a paucity of evidence, it was important to make a recommendation due to the high prevalence and importance of IPV. ANC provides an opportunity to enquire about IPV among women for whom barriers to accessing health care may exist, and also allows for the possibility for follow-up during ANC with appropriate supportive interventions, such as counselling and empowerment interventions. However, the evidence on benefits and potential harms of clinical enquiry and subsequent interventions is lacking or uncertain.
- A minimum condition for health-care providers to ask women about violence is that it must be safe to do so (i.e. the partner is not present) and that identification of IPV is followed by an appropriate response. In addition, providers must be trained to ask questions in the correct way and to respond appropriately to women who disclose violence *(86)*.
- Research on IPV is needed to answer the following questions:
  - Which are the most effective strategies for identifying, preventing and managing IPV in pregnancy?
  - Does asking routinely about violence impact on ANC attendance?
  - Can interventions targeted at partners of pregnant women prevent IPV?
- Detailed guidance on responding to IPV and sexual violence against women can be found in the 2013 WHO clinical and policy guidelines *(86)*, available at: http://www.who.int/reproductivehealth/publications/violence/9789241548595/en/

a Minimum requirements are: a protocol/standard operating procedure; training on how to ask about IPV, and on how to provide the minimum response or beyond; private setting; confidentiality ensured; system for referral in place; and time to allow for appropriate disclosure.

Summary of evidence and considerations

Effects of universal screening to detect IPV compared with no screening (usual care) (EB Table B.1.3)

The evidence on screening for IPV was derived from a Cochrane review that included two trials conducted in urban ANC settings in HICs (Canada and the USA), involving 663 pregnant women *(89)*. In one trial, 410 women were randomized before 26 weeks of gestation to a computer-based abuse assessment screening tool, with and without a provider cue sheet (giving the results of the assessment to the provider), prior to ANC consultation with a health-care provider. In the other trial (a cluster-RCT), providers administered a face-to-face screening tool that screened for 15 risk factors, including IPV, to women between 12 and 30 weeks of gestation in the intervention clusters, while women in the control clusters received usual ANC.

Low-certainty evidence from the review suggests that abuse assessment screening may identify more pregnant women with IPV than those identified through usual ANC (2 trials, 663 women; OR: 4.28, 95% CI: 1.77–10.36).

Additional considerations

- The review also pooled data on IPV screening versus no IPV screening from other health-care settings (involving pregnant and non-pregnant women), and the pooled effect estimate favoured screening to detect IPV (7 trials, 4393 women; OR: 2.35, 95% CI: 1.53–3.59).
- Another Cochrane review evaluated interventions to prevent or reduce IPV *(90)*. Uncertain evidence from one study suggests that pregnant women who receive IPV interventions (e.g. multiple counselling sessions) to prevent or reduce IPV may report fewer episodes of partner violence during pregnancy and the postpartum period (306 women; RR: 0.62, 95% CI: 0.43–0.88), but evidence on this and other outcomes is largely inconclusive.
- Most of the review evidence comes from HICs where the prevalence of women experiencing IPV in the previous 12 months ranged from 3% to 6%. However, in many settings, particularly those where economic and sociocultural factors foster a culture more permissive of violence against women, the lifetime prevalence is higher than 30%. Notably, the prevalence among young women (under 20 years old) approaches 30%, suggesting that violence commonly starts early in women's relationships *(85)*.
- Severe IPV in pregnancy (such as being beaten up, choked or burnt on purpose, being threatened with or having a weapon used against her, and sexual violence) *(85)* is more common among women who are in relationships that have also been severely abusive outside of pregnancy.
- WHO's clinical handbook on *Health care for women subjected to intimate partner violence or sexual violence* (2014) provides practical guidance on how to respond *(91)*.

Values

Please see "Women's values" in section 3.B.1: Maternal assessment: Background (p. 41).

Resources

Clinical enquiry about IPV can be conducted face-to-face or by providing women with a written or computer-based questionnaire. Although the costs of implementing these methods can vary, they might be relatively low. Subsequent management and IPV support linked to the screening intervention, however, requires sophisticated training and can therefore have significant cost implications. The GDG considered that training and resources in low-resource settings might be best targeted towards first response to IPV rather than IPV screening.

Equity

IPV is highly prevalent in many LMICs and among disadvantaged populations *(92, 93)*. Effective interventions to enquire about IPV in disadvantaged populations might help to identify those at risk of IPV-related adverse outcomes, and facilitate the provision of appropriate supportive interventions leading to improved equity. However, more evidence is needed.

Acceptability

Qualitative evidence from a range of settings on women's views of ANC suggests that pregnant women would like to be seen by a kind and supportive health-care provider who has the time to discuss issues of this nature in a private setting (high confidence in the evidence) *(22)*. However, evidence from LMICs suggests that women may be unlikely to respond favourably to cursory exchanges of information with providers who they sometimes perceive to be hurried, uncaring and occasionally abusive (high confidence in the evidence). In addition, some women may not appreciate enquiries of this nature, particularly those living in male-dominated, patriarchal societies, where women's financial dependence on their husbands may influence their willingness to discuss IPV, especially if the health professional is male *(22)*.

From the providers' perspective, qualitative evidence mainly from HICs suggests that providers often find it difficult to enquire about for IPV for the following reasons: they do not feel they have enough knowledge, training or time to discuss IPV in a sensitive manner; the presence of the partner acts as a barrier; they may have experienced IPV themselves; and they lack knowledge and guidance about the availability of additional support services (counselling, social work, etc.) (high confidence in the evidence). Providers highlight the midwife-led continuity of care (MLCC) model as a way of achieving a positive, trusting and empathetic relationship with pregnant women (moderate confidence in the evidence) (see Recommendation E.2, in section E: Health systems interventions to improve the utilization and quality of ANC).

Feasibility

Following IPV clinical enquiry, complex, multifaceted, culturally specific interventions are required to manage IPV, which could be challenging in many low-resource settings. However, emerging evidence from HICs shows that medium-duration empowerment counselling and advocacy/support, including a safety component, offered by trained health-care providers could be beneficial, and the feasibility of such interventions in LMIC settings needs investigation *(86).*

B.1.4: Gestational diabetes mellitus (GDM)

RECOMMENDATION B1.4: Hyperglycaemia first detected at any time during pregnancy should be classified as either gestational diabetes mellitus (GDM) or diabetes mellitus in pregnancy, according to WHO criteria.[a] *(Recommended)*

Remarks

- This recommendation has been integrated from the 2013 WHO publication *Diagnostic criteria and classification of hyperglycaemia first detected in pregnancy* (the strength of the recommendation and the quality of the evidence were not stated) *(94)*.
- WHO currently does not have a recommendation on whether or how to screen for GDM, and screening strategies for GDM are considered a priority area for research, particularly in LMICs.
- Diabetes mellitus in pregnancy differs from GDM in that the hyperglycaemia is more severe and does not resolve after pregnancy as it does with GDM.
- A systematic review of cohort studies shows that women with hyperglycaemia (diabetes mellitus and GDM) detected during pregnancy are at greater risk of adverse pregnancy outcomes, including macrosomia, pre-eclampsia/hypertensive disorders in pregnancy, and shoulder dystocia. Treatment of GDM, which usually involves a stepped approach of lifestyle changes (nutritional counselling and exercise) followed by oral blood-glucose-lowering agents or insulin if necessary, is effective in reducing these poor outcomes *(94)*.
- There are many uncertainties about the cost–effectiveness of different screening strategies, the prevalence of GDM and diabetes mellitus according to the 2013 criteria in diverse populations, and the impact of earlier diagnosis on pregnancy outcomes (see Chapter 5: Research implications) *(94)*.
- The usual window for diagnosing GDM is between 24 and 28 weeks of gestation. Risk factor screening is used in some settings as a strategy to determine the need for a 2-hour 75 g oral glucose tolerance test (OGTT). These include a BMI of greater than 30 kg/m^2, previous GDM, previous macrosomia, family history of diabetes mellitus, and ethnicity with a high prevalence of diabetes mellitus *(95)*. In addition, glycosuria on dipstick testing (2+ or above on one occasion, or 1+ on two or more occasions) may indicate undiagnosed GDM and, if this is observed, performing an OGTT could be considered *(95)*.
- The management approach for women classified with diabetes mellitus in pregnancy (i.e. severe hyperglycaemia first detected in pregnancy) usually differs from the approach for women with GDM, particularly when diagnosed early in pregnancy; however, the principles of management are similar and both require referral and increased monitoring.
- Further information and considerations related to this recommendation can be found in the 2013 WHO guideline *(94)*, available at: http://www.who.int/diabetes/publications/Hyperglycaemia_In_Pregnancy/en/

a This is not a recommendation on routine screening for hyperglycaemia in pregnancy. It has been adapted and integrated from the 2013 WHO publication *(94)*, which states that GDM should be diagnosed at any time in pregnancy if one or more of the following criteria are met:
- fasting plasma glucose 5.1–6.9 mmol/L (92–125 mg/dL)
- 1-hour plasma glucose ≥ 10.0 mmol/L (180 mg/dL) following a 75 g oral glucose load
- 2-hour plasma glucose 8.5–11.0 mmol/L (153–199 mg/dL) following a 75 g oral glucose load

Diabetes mellitus in pregnancy should be diagnosed if one or more of the following criteria are met:
- fasting plasma glucose ≥ 7.0 mmol/L (126 mg/dL)
- 2-hour plasma glucose ≥ 11.1 mmol/L (200 mg/dL) following a 75 g oral glucose load
- random plasma glucose ≥ 11.1 mmol/L (200 mg/dL) in the presence of diabetes symptoms..

B.1.5: Tobacco use

RECOMMENDATION B.1.5: Health-care providers should ask all pregnant women about their tobacco use (past and present) and exposure to second-hand smoke as early as possible in pregnancy and at every antenatal care visit. *(Recommended)*

Remarks

- This strong recommendation based on low-quality evidence has been integrated from the 2013 *WHO recommendations for the prevention and management of tobacco use and second-hand smoke exposure in pregnancy (96)*. Related recommendations from this guideline include the following:
  - Health-care providers should routinely offer advice and psychosocial interventions for tobacco cessation to all pregnant women who are either current tobacco users or recent tobacco quitters (strong recommendation based on moderate quality evidence).
  - All health-care facilities should be smoke-free to protect the health of all staff, patients and visitors, including pregnant women (strong recommendation based on low-quality evidence).
  - Health-care providers should provide pregnant women, their partners and other household members with advice and information about the risks of second-hand smoke (SHS) exposure from all forms of smoked tobacco, as well as strategies to reduce SHS in the home (strong recommendation based on low-quality evidence).
  - Health-care providers should, wherever possible, engage directly with partners and other household members to inform them of all the risks of SHS exposure to pregnant women from all forms of tobacco, and to promote reduction of exposure and offer smoking cessation support (strong recommendation based on low-quality evidence).
- Further guidance on strategies to prevent and manage tobacco use and SHS exposure can be found in the 2013 WHO recommendations *(96)*, available at: http://www.who.int/tobacco/publications/pregnancy/guidelinestobaccosmokeexposure/en/

B.1.6: Substance use

RECOMMENDATION B.1.6: Health-care providers should ask all pregnant women about their use of alcohol and other substances (past and present) as early as possible in the pregnancy and at every antenatal care visit. *(Recommended)*

Remarks

- This strong recommendation based on low-quality evidence has been integrated from the 2014 *WHO Guidelines for the identification and management of substance use and substance use disorders in pregnancy (97)*. The overarching principles of this guideline aimed to prioritize prevention, ensure access to prevention and treatment services, respect women's autonomy, provide comprehensive care, and safeguard against discrimination and stigmatization.
- The GDG responsible for the recommendation noted that asking women at every ANC visit is important as some women are more likely to report sensitive information only after a trusting relationship has been established.
- Pregnant women should be advised of the potential health risks to themselves and to their babies posed by alcohol and drug use.
- Validated screening instruments for alcohol and other substance use and substance use disorders are available (refer to Annex 3 of the 2014 guidelines *[97]*).
- Health-care providers should be prepared to intervene or refer all pregnant women who are identified as using alcohol and/or drugs (past and present).
- For women identified as being dependent on alcohol or drugs, further recommendations from the guideline include the following:
  - Health-care providers should at the earliest opportunity advise pregnant women dependent on alcohol or drugs to cease their alcohol or drug use and offer, or refer them to, detoxification services under medical supervision, where necessary and applicable (strong recommendation based on very low-quality evidence).
  - Health-care providers should offer a brief intervention to all pregnant women using alcohol or drugs (strong recommendation based on low-quality evidence).
- It was decided that despite the low-quality evidence on effects of brief psychosocial interventions, the benefit (potential reduction of alcohol and substance use) outweighed any potential harms, which were considered to be minimal.
- A brief intervention is a structured therapy of short duration (typically 5–30 minutes) offered with the aim of assisting an individual to cease or reduce use of a psychoactive substance.
- Further guidance on interventions and strategies to identify and manage substance use and substance use disorders in pregnancy can be found in the 2014 WHO guidelines *(97)*, available at: http://www.who.int/substance_abuse/publications/pregnancy_guidelines/en/

B.1.7: Human immunodeficiency virus (HIV) and syphilis

RECOMMENDATION B.1.7: In high-prevalence settings,[a] provider-initiated testing and counselling (PITC) for HIV should be considered a routine component of the package of care for pregnancy women in all antenatal care settings. In low-prevalence settings, PITC can be considered for pregnant women in antenatal care settings as a key component of the effort to eliminate mother-to-child transmission of HIV, and to integrate HIV testing with syphilis, viral or other key tests, as relevant to the setting, and to strengthen the underlying maternal and child health systems. *(Recommended)*

Remarks

- This recommendation has been integrated from the 2015 WHO *Consolidated guidelines on HIV testing services (98)* (the strength of the recommendation and the quality of the evidence were not stated).
- PITC denotes an HIV testing service that is routinely offered in a health-care facility and includes providing pre-test information and obtaining consent, with the option for individuals to decline testing. PITC has proved highly acceptable and has increased the uptake of HIV testing in LMICs *(98)*.
- The availability of HIV testing at ANC services is responsible for the high level of knowledge of HIV status among women in many countries, which has allowed women and infants to benefit from ART.
- WHO recommends that ART should be initiated in all pregnant women diagnosed with HIV at any CD4 count and continued lifelong *(99)*. This recommendation is based on evidence that shows that providing ART to all pregnant and breastfeeding women living with HIV improves individual health outcomes, prevents mother-to-child transmission of HIV, and prevents horizontal transmission of HIV from the mother to an uninfected sexual partner.
- Other recommendations relevant to ANC services from the *Consolidated guidelines on HIV testing services* include the following *(98)*:
  - On disclosure: Initiatives should be put in place to enforce privacy protection and institute policy, laws and norms that prevent discrimination and promote tolerance and acceptance of people living with HIV. This can help create environments where disclosure of HIV status is easier (strong recommendation, low-quality evidence).
  - On retesting: In settings with a generalized HIV epidemic:[b] Retest all HIV-negative pregnant women in the third trimester, during labour or postpartum because of the high risk of acquiring HIV infection during pregnancy (strength of recommendation and quality of evidence not stated).
  - On retesting: In settings with a concentrated HIV epidemic:[c] Retest HIV-negative pregnant women who are in a serodiscordant couple or from a key population group[d] (strength of recommendation and quality of evidence not stated).
  - On retesting before ART initiation: National programmes should retest all people newly and previously diagnosed with HIV before they enrol in care and initiate ART (strength of recommendation and quality of evidence not stated).
  - On testing strategies: In settings with greater than 5% HIV prevalence in the population being tested, a diagnosis of HIV-positive should be issued to people with two sequential reactive tests. In settings with less than 5% HIV prevalence in the population being tested, a diagnosis of HIV-positive should be issued to people with three sequential reactive tests (strength of recommendation and quality of evidence not stated).
  - On task shifting: Lay providers who are trained and supervised can independently conduct safe and effective HIV testing using rapid diagnostic tests (strong recommendation, moderate-quality evidence).
- Further guidance on HIV testing can be found in the 2015 WHO guidelines *(98)*, available at: http://www.who.int/hiv/pub/guidelines/hiv-testing-services/en/
- In addition, the 2015 *Guideline on when to start antiretroviral therapy and on pre-exposure prophylaxis for HIV (99)* is available at: http://www.who.int/hiv/pub/guidelines/earlyrelease-arv/en/
- To prevent mother-to-child transmission of syphilis, all pregnant women should be screened for syphilis at the first ANC visit in the first trimester and again in the third trimester of pregnancy. For further guidance

on screening, please refer to the 2006 WHO publication *Prevention of mother-to-child transmission of syphilis (100)*, available at: http://www.who.int/reproductivehealth/publications/maternal_perinatal_health/prevention_mtct_syphilis.pdf

- The latest (2016) WHO guidelines on the treatment of chlamydia, gonorrhoea and syphilis, and on the prevention of sexual transmission of Zika virus *(101–104)*, are available at: http://www.who.int/reproductivehealth/publications/rtis/clinical/en/

a High-prevalence settings are defined in the 2015 WHO publication *Consolidated guidelines on HIV testing services* as settings with greater than 5% HIV prevalence in the population being tested. Low-prevalence settings are settings with less than 5% HIV prevalence in the population being tested *(98)*.
b A generalized HIV epidemic is when HIV is firmly established in the general population. Numerical proxy: HIV prevalence is consistently over 1% in pregnant women attending antenatal clinics *(98)*.
c A concentrated HIV epidemic is when HIV has spread rapidly in a defined subpopulation (or key population, see next footnote) but is not well established in the general population *(98)*.
d Key populations are defined in the 2015 WHO guidelines as the following groups: men who have sex with men, people in prison or other closed settings, people who inject drugs, sex workers and transgender people *(98)*.

B.1.8: Tuberculosis (TB)

RECOMMENDATION B.1.8: In settings where the tuberculosis (TB) prevalence in the general population is 100/100 000 population or higher, systematic screening for active TB should be considered for pregnant women as part of antenatal care. *(Context-specific recommendation)*

Remarks

- This recommendation has been adapted and integrated from the 2013 WHO publication *Systematic screening for active tuberculosis: principles and recommendations*, where it was considered a conditional recommendation based on very low-quality evidence *(105)*.
- Systematic screening is defined as the systematic identification of people with suspected active TB in a predetermined target group, using tests, examinations or other procedures that can be applied rapidly. Options for initial screening include screening for symptoms (either for cough lasting longer than two weeks, or any symptoms compatible with TB, including a cough of any duration, haemoptysis, weight loss, fever or night sweats) or screening with chest radiography. The use of chest radiography in pregnant women poses no significant risk but the national guidelines for the use of radiography during pregnancy should be followed *(105)*.
- Before screening is initiated, high-quality TB diagnosis, treatment, care, management and support should be in place, and there should be the capacity to scale these up further to match the anticipated rise in case detection that may occur as a result of screening.
- The panel responsible for making this recommendation noted that it may not be possible to implement it in resource-constrained settings.
- Other recommendations relevant to ANC services from the same publication include the following *(105)*:
  - Household contacts and other close contacts should be systematically screened for TB (strong recommendation, very low-quality evidence).
  - People living with HIV should be systematically screened for active TB at each visit to a health-care facility (strong recommendation, very low-quality evidence).
  - Systematic screening for active TB may be considered also for other subpopulations that have very poor access to health care, such as people living in urban slums, homeless people, people living in remote areas with poor access to health care, and other vulnerable or marginalized groups including some indigenous populations, migrants and refugees (conditional recommendation, very low-quality evidence).
- TB increases the risk of preterm birth, perinatal death and other pregnancy complications. Initiating TB treatment early is associated with better maternal and infant outcomes than late initiation *(105)*.
- To better understand the local burden of TB in pregnancy, health systems may benefit from capturing pregnancy status in registers that track TB screening and treatment.
- Further information and considerations related to this recommendation can be found in the 2013 WHO recommendations *(105)*, available at: http://www.who.int/tb/tbscreening/en/